Class 7: Drude Model: Electrical Conductivity

We are familiar with Ohm's law written as an equation in the form:

$$V = IR$$

Since resistance experienced by electrons arises due to collisions with other electrons, it is not surprising that the resistance increases as the length of the conductor increases, since that increases the number of collisions that the electron will experience. Similarly, the resistance is seen to decrease as the cross sectional area of the conductor increases, since the increased area provides parallel paths for the electron to travel.

Therefore, if the resistivity of the sample is ρ, the resistance of the sample is given by

$$R = \frac{\rho L}{A}$$

Substituting into Ohm's law, we get:

$$V = I\frac{\rho L}{A}$$

Rearranging:

$$\frac{V}{L} = \frac{I}{A}\rho$$

Since

$$\frac{V}{L} = \varepsilon$$

where ε is the electric field,
and

$$\frac{I}{A} = J$$

where J is the current density,

$$\frac{V}{L} = \frac{I}{A}\rho$$

Becomes:

$$\varepsilon = J\rho$$

Since the resistivity ρ is the inverse of conductivity σ, we can rewrite the above as:

$$J = \varepsilon\sigma$$

The above equation is simply another way to state the Ohm's law. We will use the above expression in the discussion that follows since we will be able to more readily relate it to the phenomena in the material.

We will now examine how electrons move when they experience an electric field. In this process we will get a relation between J and ε, and this will have another quantity which we will then associate with the conductivity σ

In this exercise, we will use the free electron theory to arrive at an expression for σ of a metal. This approach is credited to Drude and Lorenz and treats the free electrons in a solid in a manner similar to atoms of an ideal gas.

Assume a 1 Dimensional sample of length L, across which we apply a potential difference V.

The field generated due to this is given by:

$$\varepsilon = \frac{V}{L}$$

The force F, experienced by the electrons is given by:

$$F = \varepsilon e$$

Where e, is the charge of an electron

Dimensionally, we can check that the above is correct since the RHS has the dimensions Coloumb Volt / meter, which is Joules/m, which is equal to Newtons.

The force F, is mass times acceleration, therefore

$$F = m\frac{dv}{dt} = \varepsilon e$$

Since the applied field, charge of the electron, and mass of the electron are constant, the above equation implies that the electrons will experience constant acceleration. Constant acceleration will result in a continuously increasing velocity of the electrons and hence a continuously increasing current. In reality this is not observed – the current reaches a steady state value.

As the electrons move within the conductor, they collide with other electrons present, and the faster they move, the more likely it is that they will undergo such a collision. Therefore there is some form of a general resistive term that becomes more prevalent as the velocity of the electron increases.

In the assumptions that we listed at the end of the last class, in order to extend the ideal gas theory to free electrons, we mentioned the need to introduce an 'approximate term' that accounts for the interaction between the electrons and between the electrons and the ionic cores as a whole. We will now introduce this approximate term.

We will list the resistive term as

$$\gamma v$$

Such that as v increases, the resistive force increases.

Therefore, the net force experienced by the electron can now be written as:

$$m\frac{dv}{dt} = \varepsilon e - \gamma v$$

In the above equation, the accelerating force is constant, while the resistive force increases with the velocity of the electrons. Therefore, with the passage of time, as the velocity of the electrons increases, the net force on the electrons drops to zero, and the electrons attain a constant final velocity v_f, known as the drift velocity.

$$\varepsilon e - \gamma v_f = 0$$

Rearranging, we get:

$$\gamma = \frac{\varepsilon e}{v_f}$$

Substituting back we get:

$$m\frac{dv}{dt} = \varepsilon e - \frac{\varepsilon e}{v_f} v$$

Here, the velocity v is a variable, while v_f is a constant. It is relevant to note here, that even in the absence of an applied electric field, the free electrons in a metal are in constant random motion at any temperature above absolute zero. However, in the absence of an applied field, the net velocity is zero. This is consistent with the fact that we do not see current in the absence of an applied potential or electric field. When an electric field is applied, a net velocity consistent with the direction of the applied field, is observed. Therefore the net velocity is a quantity that is initially zero, and then increases in a given direction in response to an applied field in that direction.

Incidentally the ratio $\frac{v_f}{\varepsilon}$, represents the velocity attained by the species per unit driving force and is referred to as mobility, μ, of the species. Mobility is a more general concept and appears in the treatment of other phenomena such as diffusion.

We can now rearrange the above equation as follows:

$$m\frac{dv}{dt} = \varepsilon e\left[1 - \frac{v}{v_f}\right]$$

Or

$$\frac{dv}{\left[1 - \frac{v}{v_f}\right]} = \frac{\varepsilon e}{m} dt$$

The above can be integrated with v varying from 0 to v, while t varies from 0 to t

$$\int_0^v \frac{dv}{\left[1 - \frac{v}{v_f}\right]} = \int_0^t \frac{\varepsilon e}{m} dt$$

Therefore

$$-v_f \ln\left[1 - \frac{v}{v_f}\right] = \frac{\varepsilon e}{m} t + constant$$

LHS of the equation above evaluates to zero, when $v_f = 0$. And $v_f = 0$ when $t = 0$ therefore the $constant$ in the equation above is also zero.

Rearranging, we get:

$$\ln\left[1-\frac{v}{v_f}\right] = -\frac{\varepsilon e}{m v_f} t$$

The term $\frac{\varepsilon e}{m v_f}$ has the dimensions of (time)$^{-1}$ since it is acceleration (force/mass) divided by velocity. Let us denote the time indicated by this term as τ and examine what τ can be associated with. Assuming an electron starts with a velocity zero, and is subject to an acceleration $\frac{\varepsilon e}{m}$, then from Newton's first law, the velocity it will attain after time τ is as follows:

$$v = 0 + \frac{\varepsilon e}{m}\tau$$

Looking at the expression for τ, we see that $v = v_f$, therefore

$$v_f = 0 + \frac{\varepsilon e}{m}\tau$$

Therefore we can associate τ with the time taken to attain v_f if there is no change in the acceleration within this time, implying that there are no collisions within this time. We can therefore consider the time τ as the mean time between collisions. In reality the typical electron may encounter a few collisions before it reaches v_f, we are looking at one possible scenario in the discussion above.

Therefore, we can now write the equation as below:

$$\ln\left[1-\frac{v}{v_f}\right] = -\frac{t}{\tau}$$

This equation can be rearranged as follows:

$$1-\frac{v}{v_f} = exp\left(-\frac{t}{\tau}\right)$$

$$\frac{v}{v_f} = 1 - exp\left(-\frac{t}{\tau}\right)$$

$$v = v_f\left[1 - exp\left(-\frac{t}{\tau}\right)\right]$$

This implies that at $t = 0, v = 0$ and at $t = \infty, v = v_f$

A plot of this equation is shown in Figure 7.1 below. τ is of the order of 10^{-14} seconds. Therefore, when we switch on a circuit, it takes approximately $5X10^{-14}$ seconds for the current to stabilize, which is virtually instantaneous from our perspective.

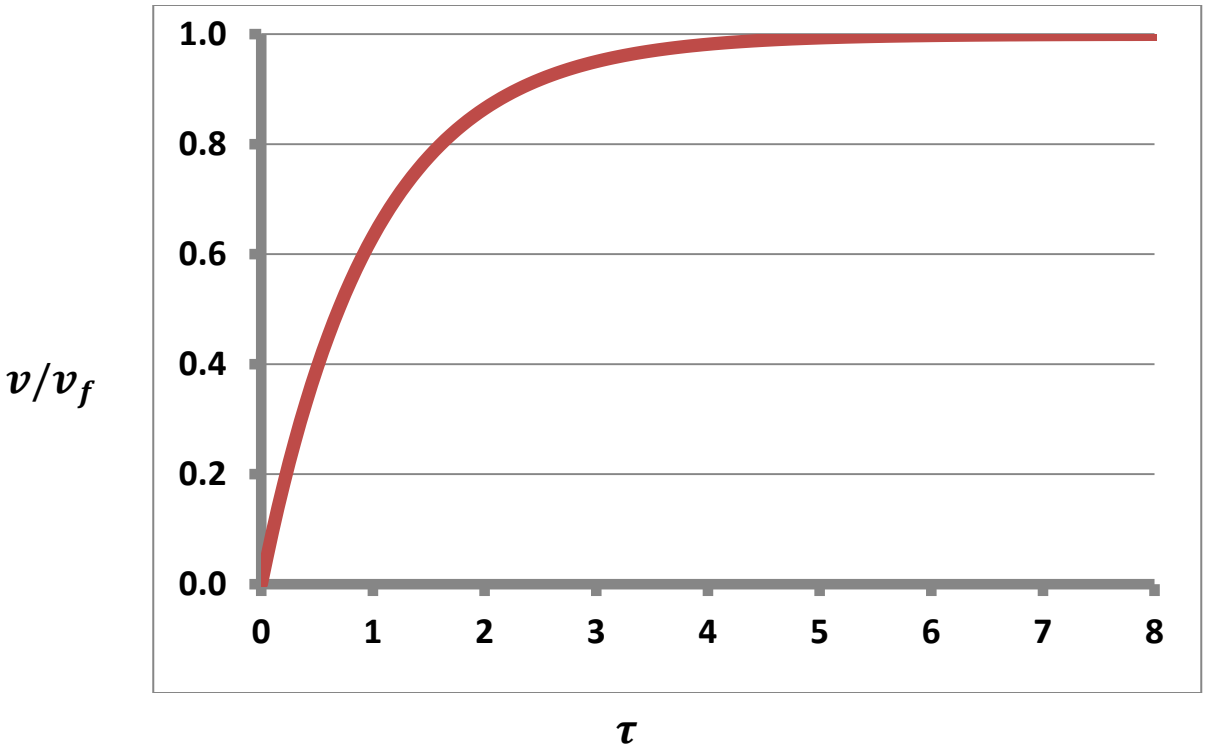


Figure 7.1: Variation of v/v_f as a function of τ

Looking at the system from a different perspective, it is of interest to identify the charge carriers and to identify the number of charge carriers that are moving, since this is what causes the current. In the system we are examining, electrons are the charge carriers and the number of free electrons per unit volume, N_f, is the quantity of interest with respect to the current that is flowing in the system. The speed with which the electrons move is the drift velocity, v_f.

Therefore, the current density J is given by:

$$J = N_f v_f e$$

The RHS of the equation has the units Coloumbs per square meter per second, which is the units of the current density.

As indicated earlier,

$$\frac{\varepsilon e}{m v_f} = \frac{1}{\tau}$$

Therefore

$$\frac{\tau \varepsilon e}{m} = v_f$$

Substituting in the expression for current density, we have

$$J = N_f \frac{\tau \varepsilon e}{m} e$$

Or

$$J = \frac{N_f e^2 \tau}{m} \varepsilon$$

Comparing with

$$J = \varepsilon \sigma$$

We have

$$\sigma = \frac{N_f e^2 \tau}{m}$$

In this equation,

$$N_f = 10^{28}\ m^{-3}$$
$$e = 1.6 \times 10^{-19}\ C$$
$$\tau = 10^{-14}\ s$$
$$m = 9.1 \times 10^{-31}\ kg$$

Therefore

$\sigma = 3 \times 10^6$ $(\Omega m)^{-1}$

The above is a reasonably good prediction since metallic systems display conductivity values of the order of 10^7 $(\Omega m)^{-1}$. As indicated in our earlier discussions, we are more interested in getting the correct order of magnitude for the predictions, rather than the exact value.

Here we have assumed a unit valency for the metal. The valency we assume will affect the N_f value since higher the valency, greater will be the N_f value. At the same time, if N_f increases, we can reasonably expect τ to decrease since the electrons will now be more likely to collide with each other.

Correct prediction of the electronic conductivity of metallic systems is therefore a success for the free electron theory of metals or the Drude model. In the next class we will use the same set of assumptions and try to predict the thermal conductivity of metallic systems. We will also examine how well the Drude model predicts the relationship between electrical and thermal conductivities of metals.